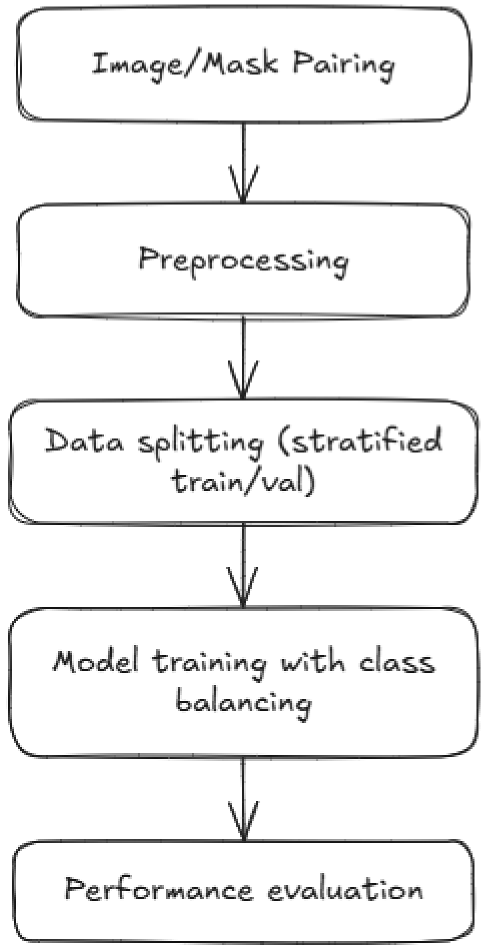
Team 16

Project Pipeline

Objective: Build a deep learning model to segment cloud coverage in satellite imagery.

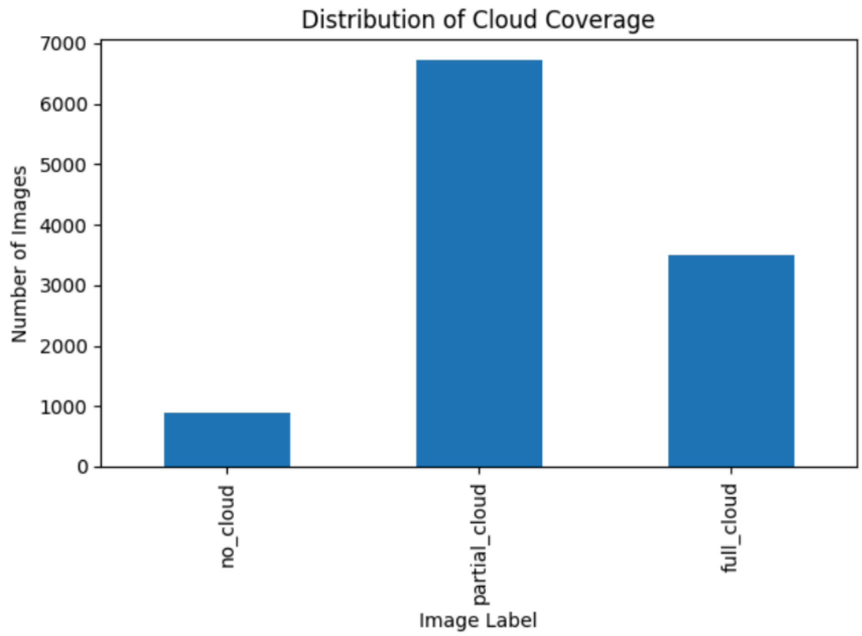


Preprocessing Module

- Pairing images with masks by filename.
- Cloud fraction statistics (from EDA) are used to understand label distribution.
- Stratified splitting based on cloud type: no_cloud, partial_cloud, full_cloud.
- Normalization/standardization: We normalized each band in the image.

Exploratory Data Analysis

- Summary statistics for cloud coverage:
    - count 11123.000000
    - mean 0.623952
    - std 0.376043
    - min 0.000000
    - 25% 0.271769
    - 50% 0.731297
    - 75% 1.000000
    - max 1.000000



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- Insights:
    - We can see that the images with no cloud are under represented in the dataset.
    - This data imbalance will limit the model's ability to detect no cloud images.

Model Selection & Training Module

- Chosen Model: U-Net with 4 input channels (Red, Green, Blue, Infrared) and 1 output channel (cloud segmentation).

- Loss: Combination of Binary Cross-Entropy (BCE) and Dice Loss.
- Optimizer: Adam.
- Scheduler: Cosine Annealing LR.
- Class balancing: Used WeightedRandomSampler based on cloud class frequencies.
- Checkpointing: Saved the best model based on validation loss.

Performance Analysis Module

Metrics used:

- BCE Loss
- Dice Loss

Tracked the train and validation loss using tqdm to detect any issues during the training progress.

Additional Developed Modules

To explore a more lightweight, classical machine learning approach, we also experimented with training a pixel-wise classifier using SGDClassifier from scikit-learn, treating each pixel as an independent sample.

Experiment Details:

- Input Features: (Red Green Blue Infrared) values per pixel.
- Target: Binary mask value (cloud = 1, no cloud = 0).
- Model: SGDClassifier with log_loss (i.e., logistic regression) and partial_fit.
- Training Approach:
- For each batch:
    - Flatten the batch into a set of pixel-level samples.
    - Train incrementally using partial_fit.
    - Classes were explicitly passed during the first call to partial_fit to handle class imbalance.
- Validation Accuracy: achieved approximately 0.7949
- Strengths: very fast training, low memory usage.
- Limitations:
    - Ignored spatial relationships between pixels.
    - Much lower performance compared to U-Net because of lack of spatial context.

Enhancements and Future Work

- Try different more advanced architectures.
- Investigate use of more channels that might be more helpful in this problem.
- Conduct full cross-validation instead of single split.

Team Members

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